

On Data-Driven Control of Linear Systems with Quantized Data

Controller design and stabilization with finite-valued state measurements

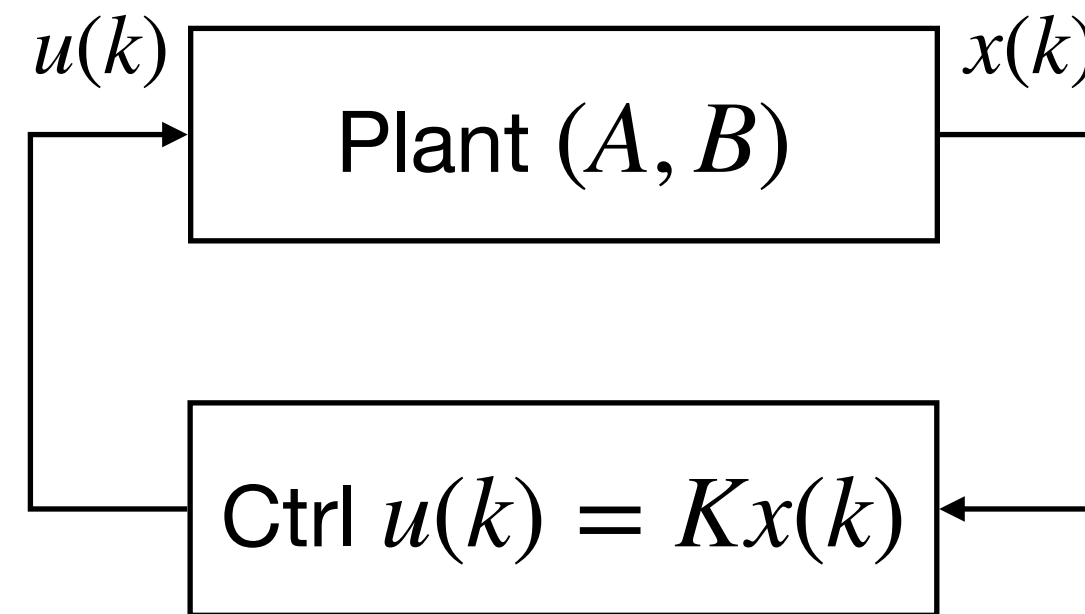
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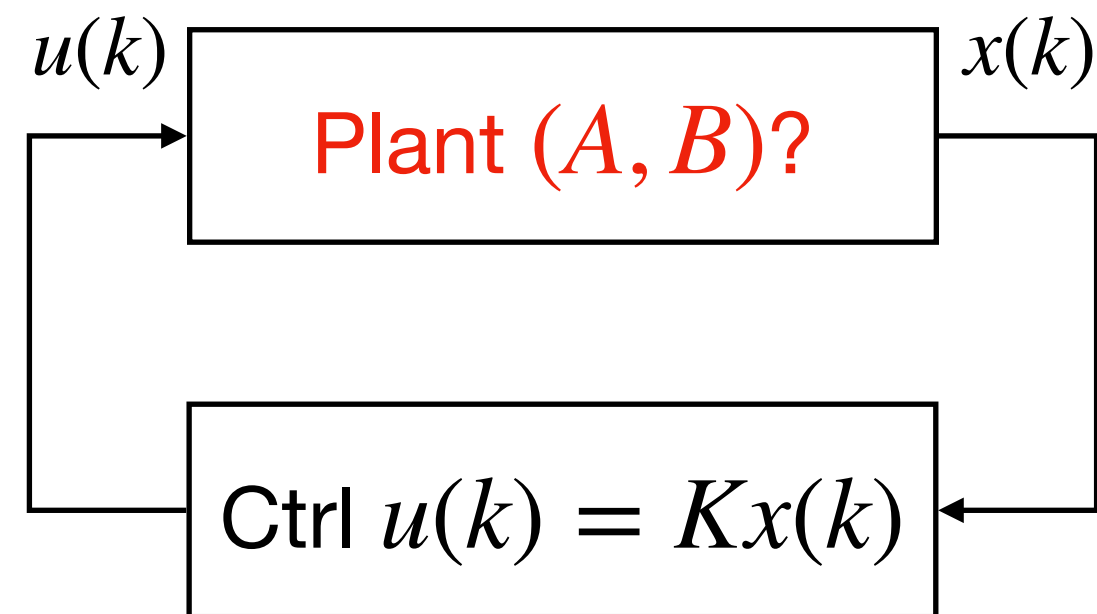
The 62nd Allerton Conference on Communication, Control, and Computing

Linear state-feedback control



- **Design:** $(A, B) \rightarrow$ controller design $\rightarrow K$
- **Implementation:** $x(k) \rightarrow$ controller $\rightarrow u(k) = Kx(k)$
- **This work:** unknown plant + quantized state measurements

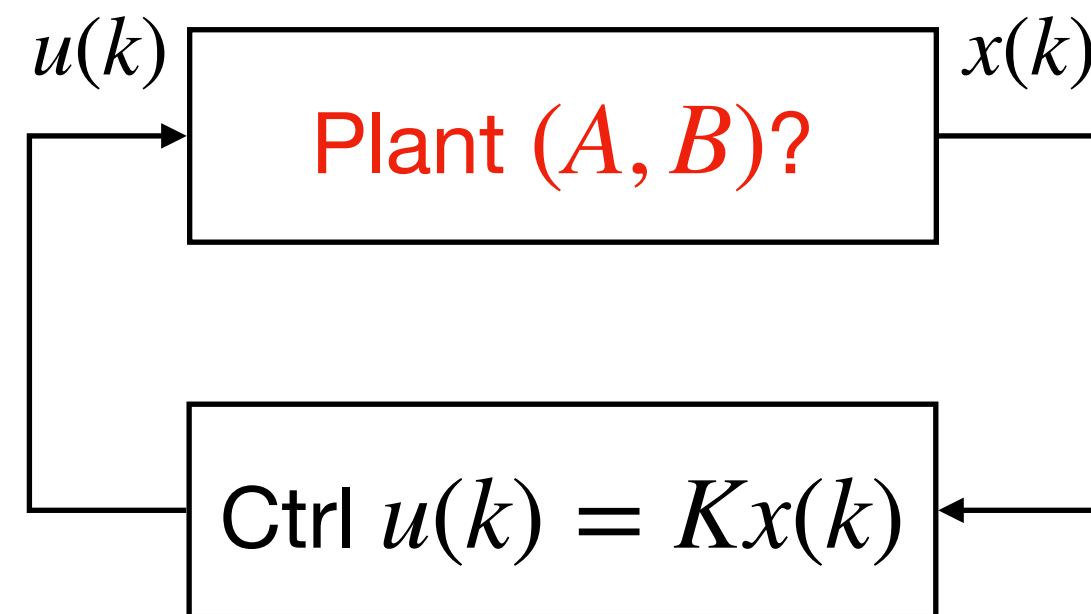
Related approaches and gap



Data-driven control

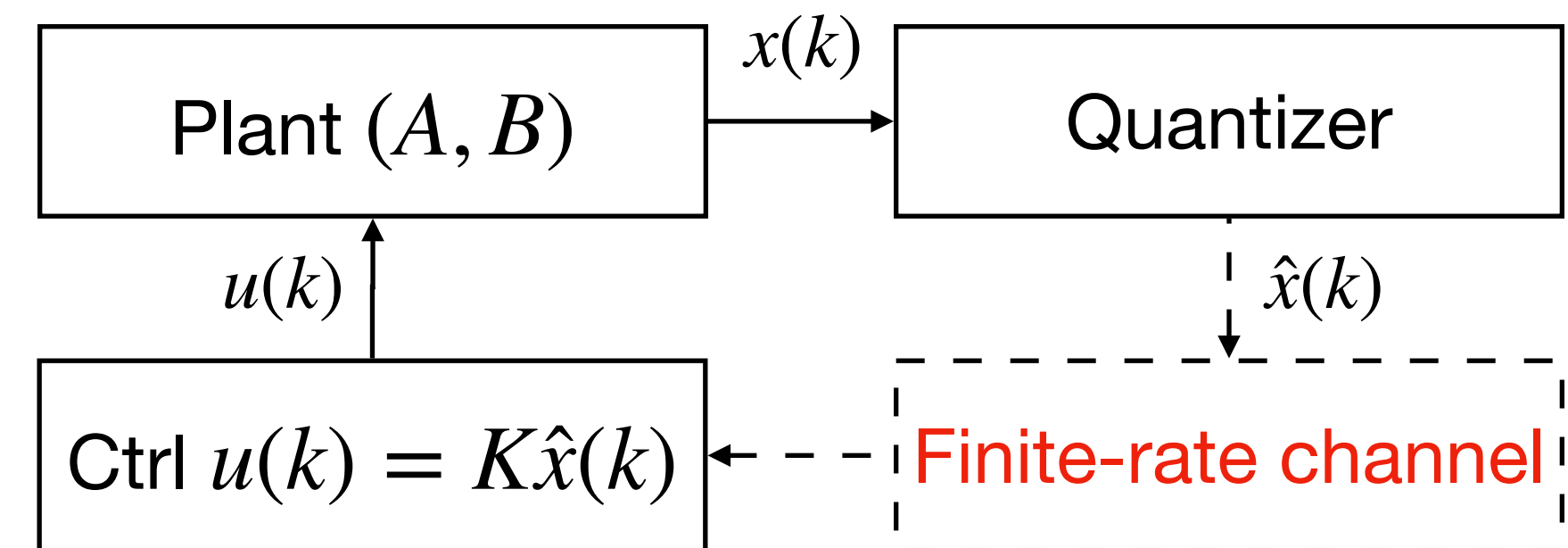
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- No explicit model identification
- Typically assumes unquantized state data

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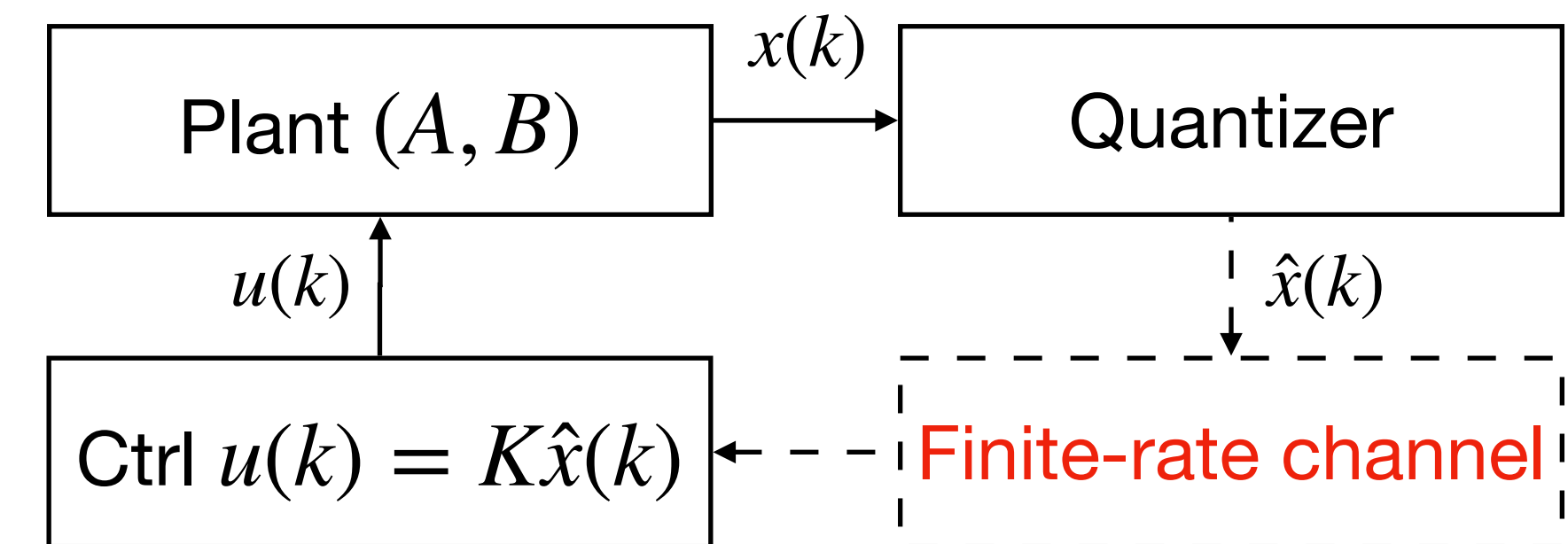
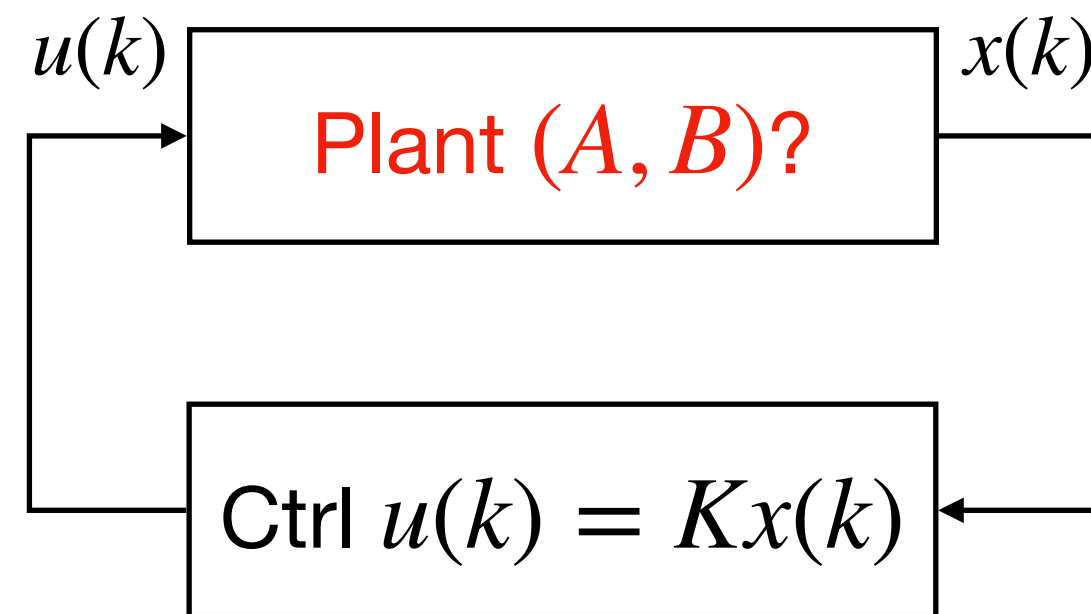
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Control under communication constraints

- Communication channel with finite data rate
- Quantized state measurements
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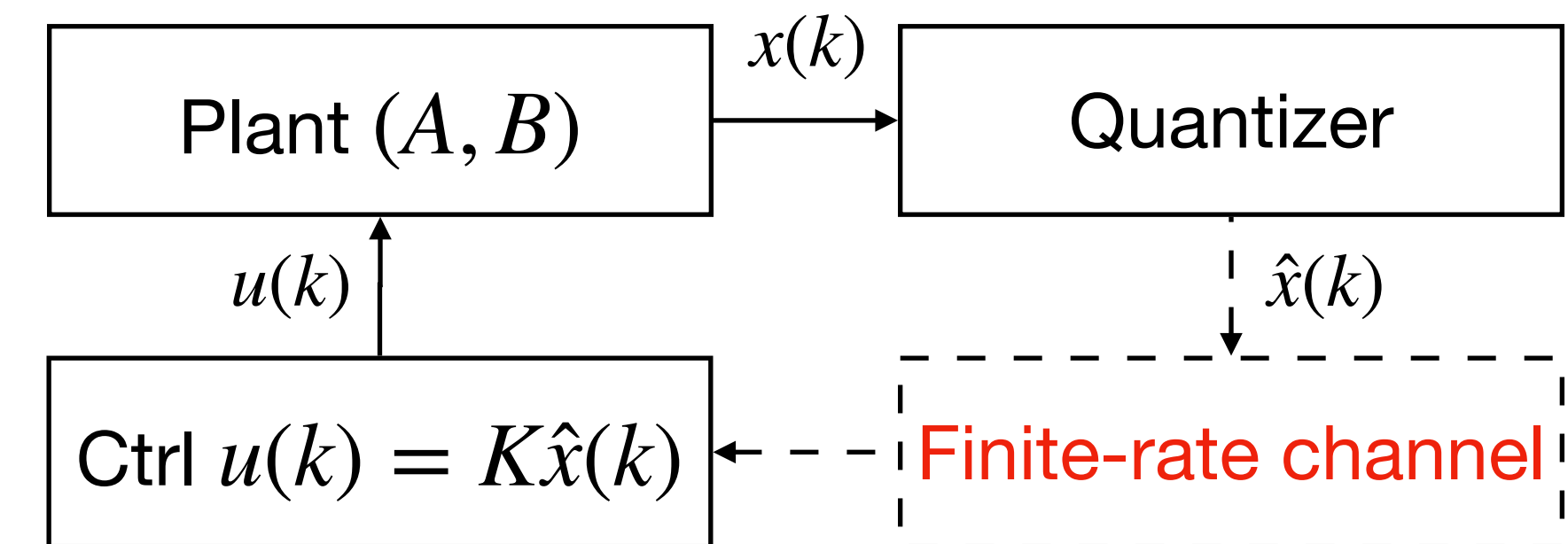
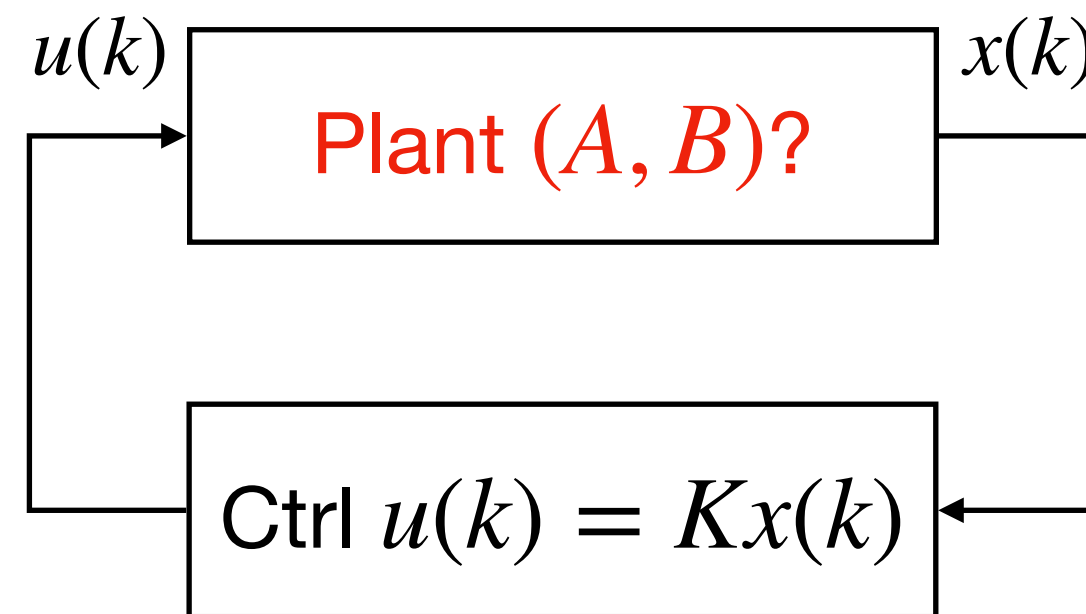
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Recent data-driven control approaches with quantization

- Controller design often relies at least partly on **unquantized data**
- Feedback typically uses **infinite-level quantizers**

Related approaches and gap



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Recent data-driven

- Controller design
- Feedback type

Can we perform both controller design and stabilization using only finite-valued quantized state measurements?

Problem formulation

$$x(k+1) = Ax(k) + Bu(k), \quad k \in \mathbb{Z}_{\geq 0}$$

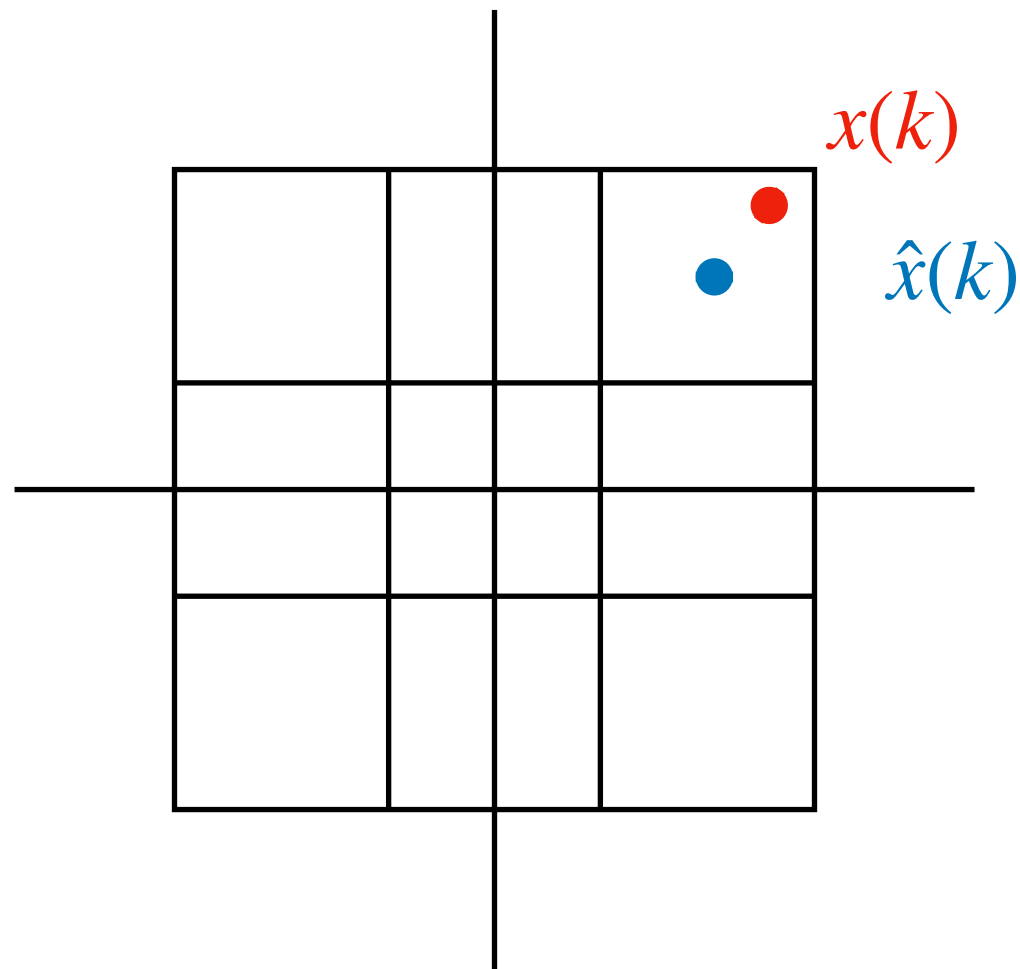
- State $x \in \mathbb{R}^n$, control input $u \in \mathbb{R}^m$
- A, B unknown, $\|A\| \leq \alpha$, $\|B\| \leq \beta$

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Quantized measurement $\hat{x}(k) = q_k(x(k))$, $q_k : \mathbb{R}^n \rightarrow \mathcal{Q}_k$, $|\mathcal{Q}_k| < \infty$



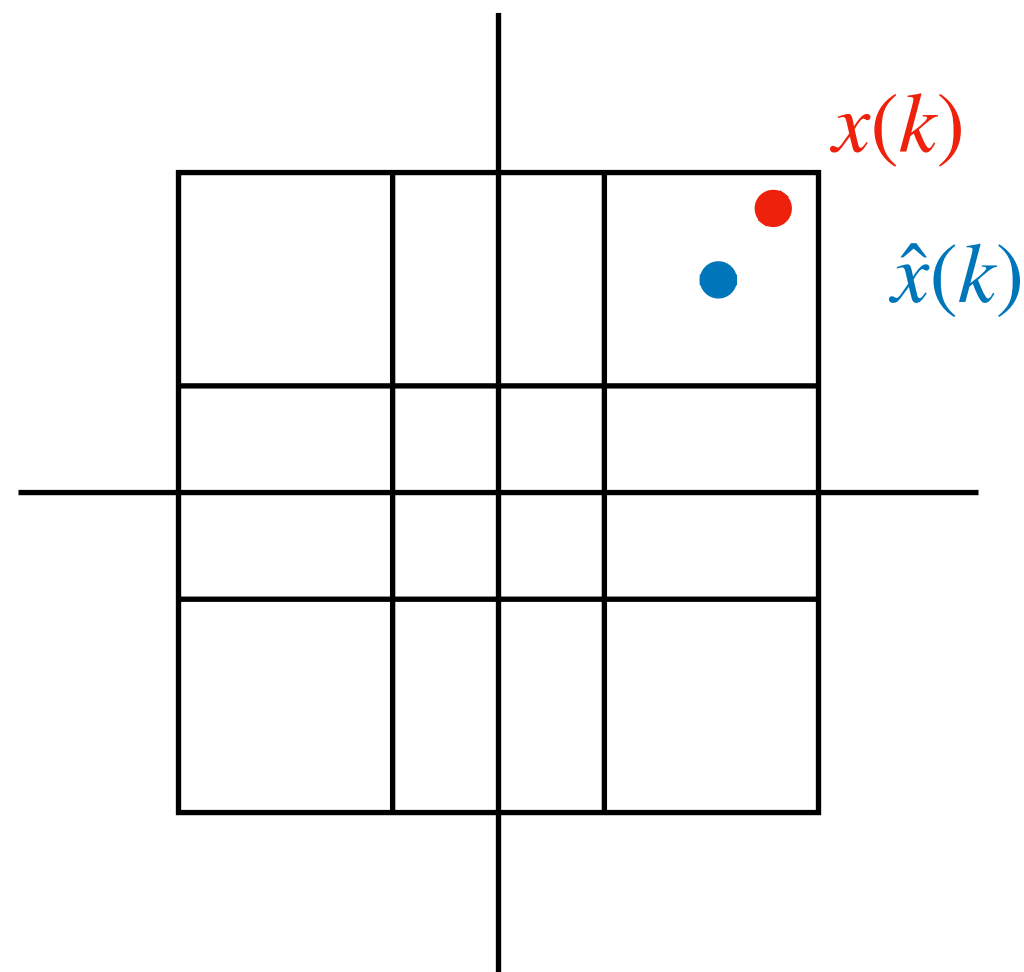
- Range $\mathcal{R}(E_k) = \{x : \|x\|_\infty \leq E_k\}$
- N_k cells per dimension
- Error $\|d(k)\|_\infty = \|\hat{x}(k) - x(k)\|_\infty \leq \frac{E_k}{N_k}$

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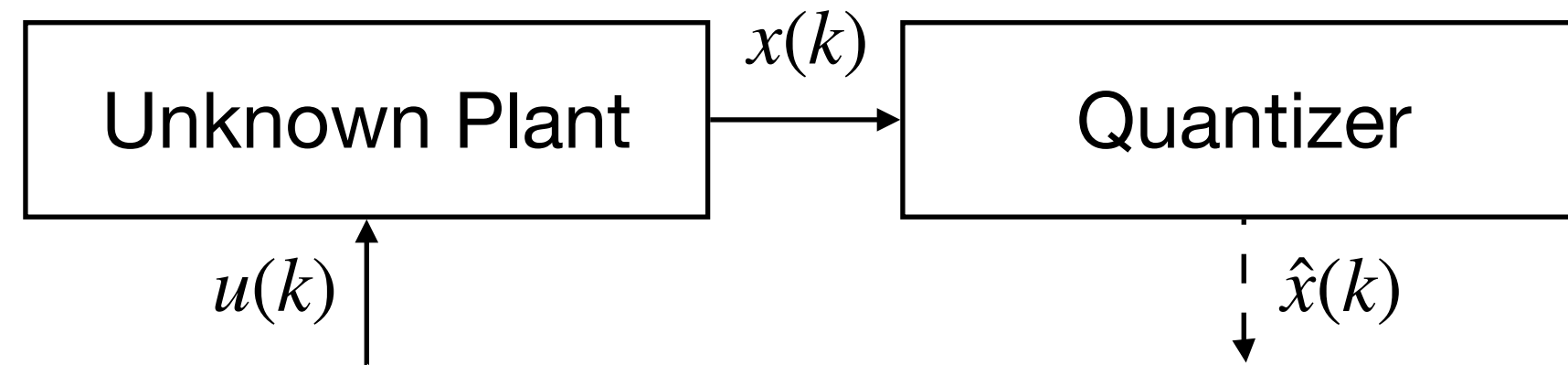
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Objective: Design quantizers and controller using only $u(k)$ and $\hat{x}(k)$ such that $x(k) \rightarrow 0$.

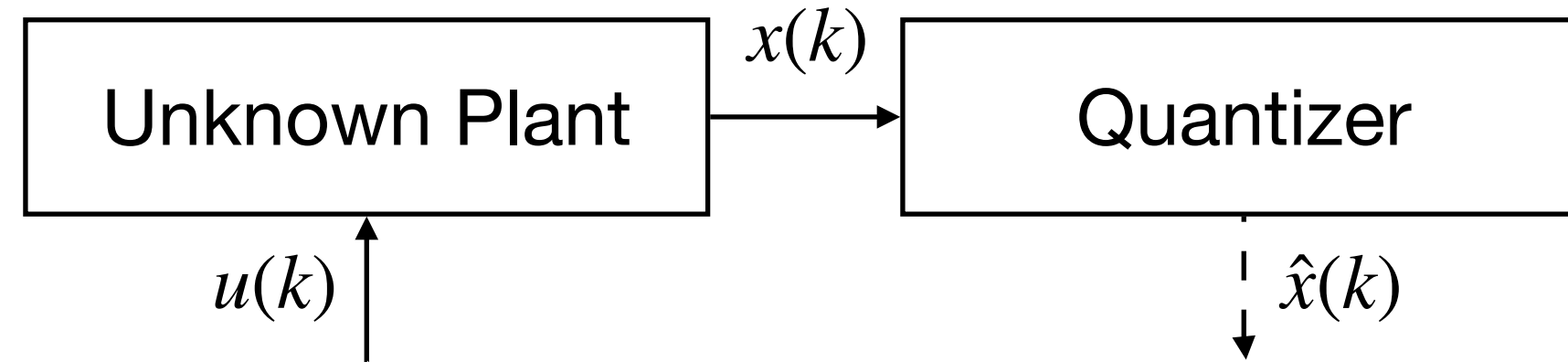
Proposed two-stage framework



Stage 1: Controller design

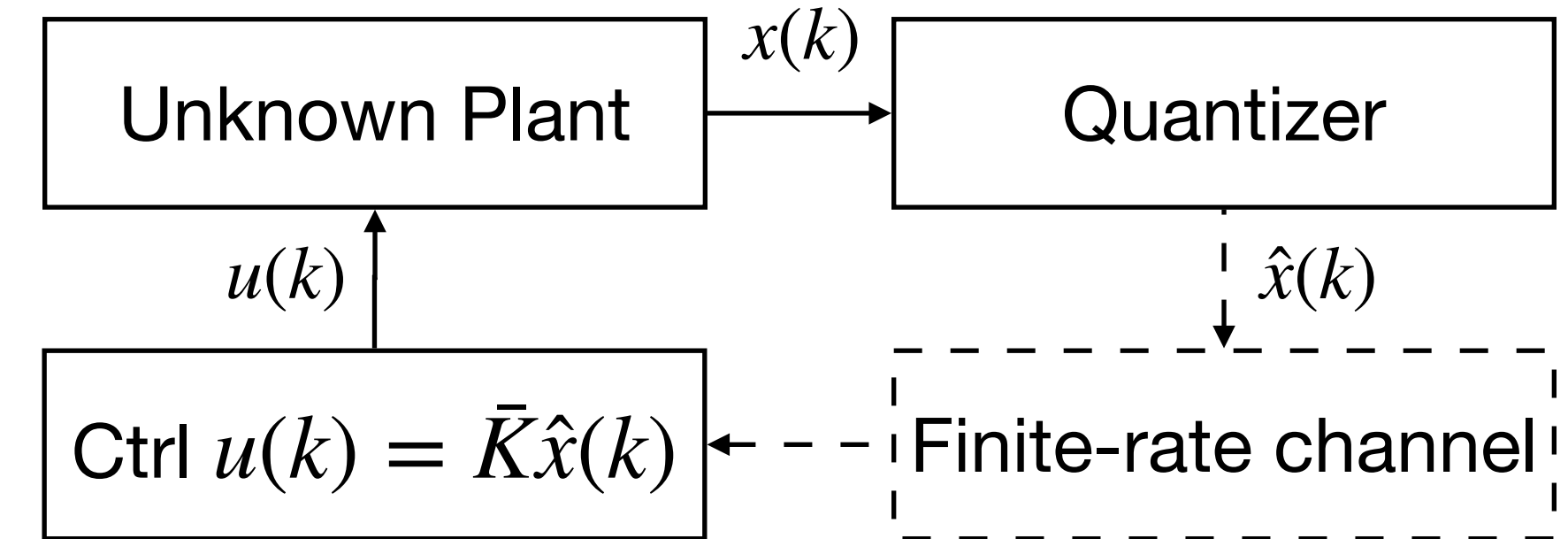
- Persistently exciting (PE) input $u(k)$
- Quantizer with prescribed error bound δ
- Quantized data $\hat{x}(k)$
- Stabilizing controller $\bar{K}$ via semidefinite program (SDP)

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Stage 2: Stabilization

- Quantized feedback $u(k) = \bar{K}\hat{x}(k)$
- Quantizer with fixed N_k and shrinking E_k
- Exponential convergence for $x(k)$

Controller design: Quantized data collection

$$x(k+1) = Ax(k) + Bu(k), \quad k = 0, 1, \dots, T$$

- $\|x(0)\|_{\infty} \leq E_0$
- Bounded input $\|u(k)\| \leq \bar{u}$, PE of order $n+1$
- $T \geq m(n+1) + n$

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Quantizer update

- $\|x(k)\|_\infty \leq E_k, \quad \|\hat{x}(k) - x(k)\|_\infty \leq \delta$
- $N_k = \left\lceil \frac{E_k}{\delta} \right\rceil, \quad E_{k+1} = \alpha \sqrt{n} E_k + \beta \bar{u}$

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Less conservative: $E_{k+1} = \alpha(\|\hat{x}(k)\| + \sqrt{n}\delta) + \beta\bar{u}$

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- E_k, N_k are synchronized by their update rules $\rightarrow$ only finite-valued $\hat{x}(k)$ is transmitted.

Controller design: Quantized data collection

Data matrices

- $\hat{X}_0 := [\hat{x}(0) \quad \hat{x}(1) \quad \cdots \quad \hat{x}(T-1)], \quad \hat{X}_1 := [\hat{x}(1) \quad \hat{x}(2) \quad \cdots \quad \hat{x}(T)]$
- $U_0 := [u(0) \quad u(1) \quad \cdots \quad u(T-1)]$
- They satisfy

$$\hat{X}_1 = A\hat{X}_0 + BU_0 + \Delta_0, \quad \|\Delta_0\| \leq \sqrt{nT}(\alpha + 1)\delta =: \hat{\delta}$$

Controller design: Controller synthesis

Quantized SDP

$$\min_{(Q,P)} \text{tr}(P) + \text{tr}(U_0 Q P^{-1} Q^\top U_0^\top) + \omega \text{tr}(Q P^{-1} Q^\top)$$

subject to

$$\hat{X}_1 Q P^{-1} Q^\top \hat{X}_1^\top - P + I \preceq 0,$$

$$P \succeq I,$$

$$\hat{X}_0 Q = P$$

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subject to

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$$P \succeq I,$$

$$\hat{X}_0 Q = P$$

- For a solution $(\bar{Q}, \bar{P})$, construct feedback gain $\bar{K} = U_0 \bar{Q} \bar{P}^{-1}$

Controller design: Guarantees

Theorem 1 (Verifiable certificate)

- For any feasible solution, compute $\eta_1 := 1 - \hat{\delta}^2 \|\bar{Q}\bar{P}^{-1}\bar{Q}^\top\| - 2\hat{\delta}\|\hat{X}_1\bar{Q}\bar{P}^{-1}\bar{Q}^\top\|$
- $\eta_1 > 0 \Rightarrow A + B\bar{K}$ is Schur stable

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Theorem 2 (Existence guarantee)

- Suppose controllable (A, B) and PE input.
- There exists $\delta^* > 0$ such that $\delta < \delta^* \implies$ feasible SDP & $\eta_1 > 0$ for every optimal solution

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1. $\eta_1 > 0$ provides a verifiable certificate ensuring the obtained feedback gain is stabilizing.
2. Sufficiently fine quantization guarantees the verifiable condition.

Stabilization: Quantizer update

$$x(k+1) = Ax(k) + Bu(k), \quad k \geq T+1$$

- Quantized state feedback $u(k) = \bar{K}\hat{x}(k)$

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Quantizer update:

- $N_k = N_s, \quad E_{k+1} = \mu E_k$
- Lyapunov function $V(x) = x^\top \bar{P}^{-1}x$, with **sublevel sets** $\mathcal{S}(E) = \{x : V(x) \leq E^2/\bar{\lambda}(\bar{P})\} \subset \mathcal{R}(E)$

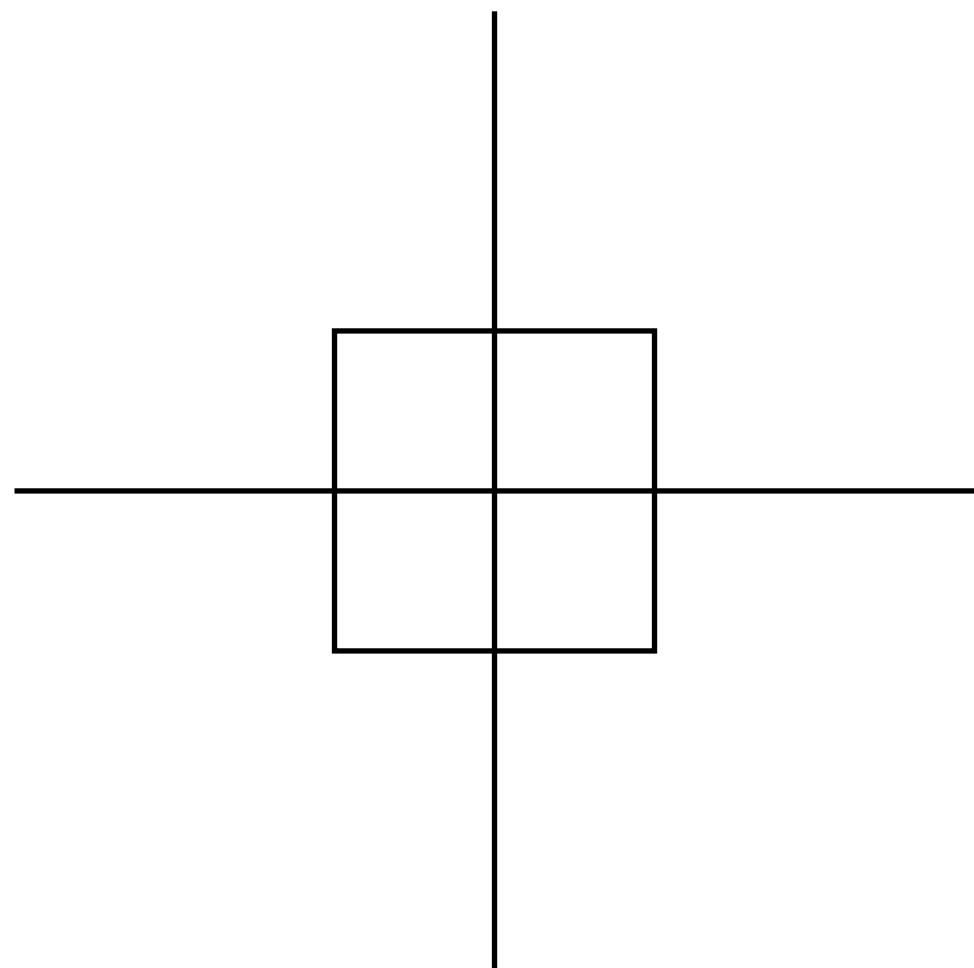
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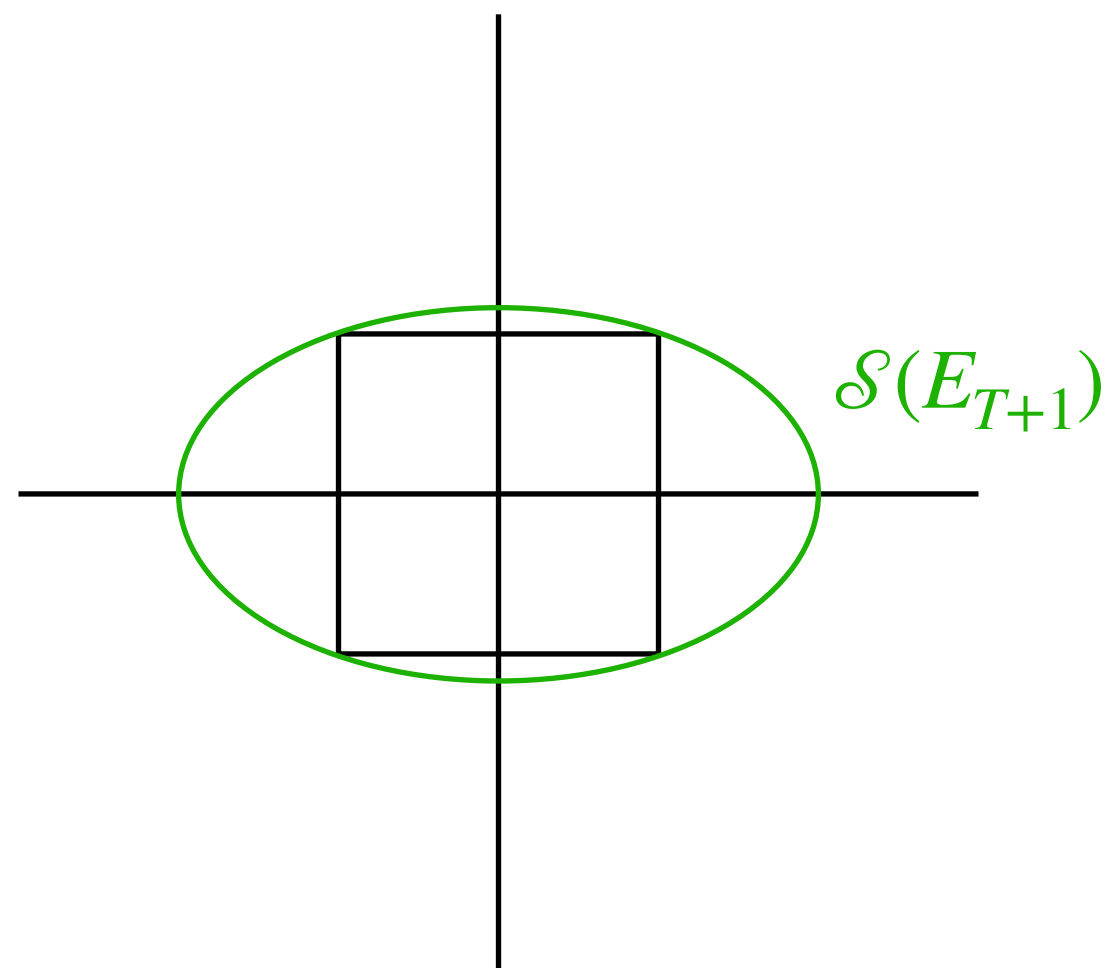
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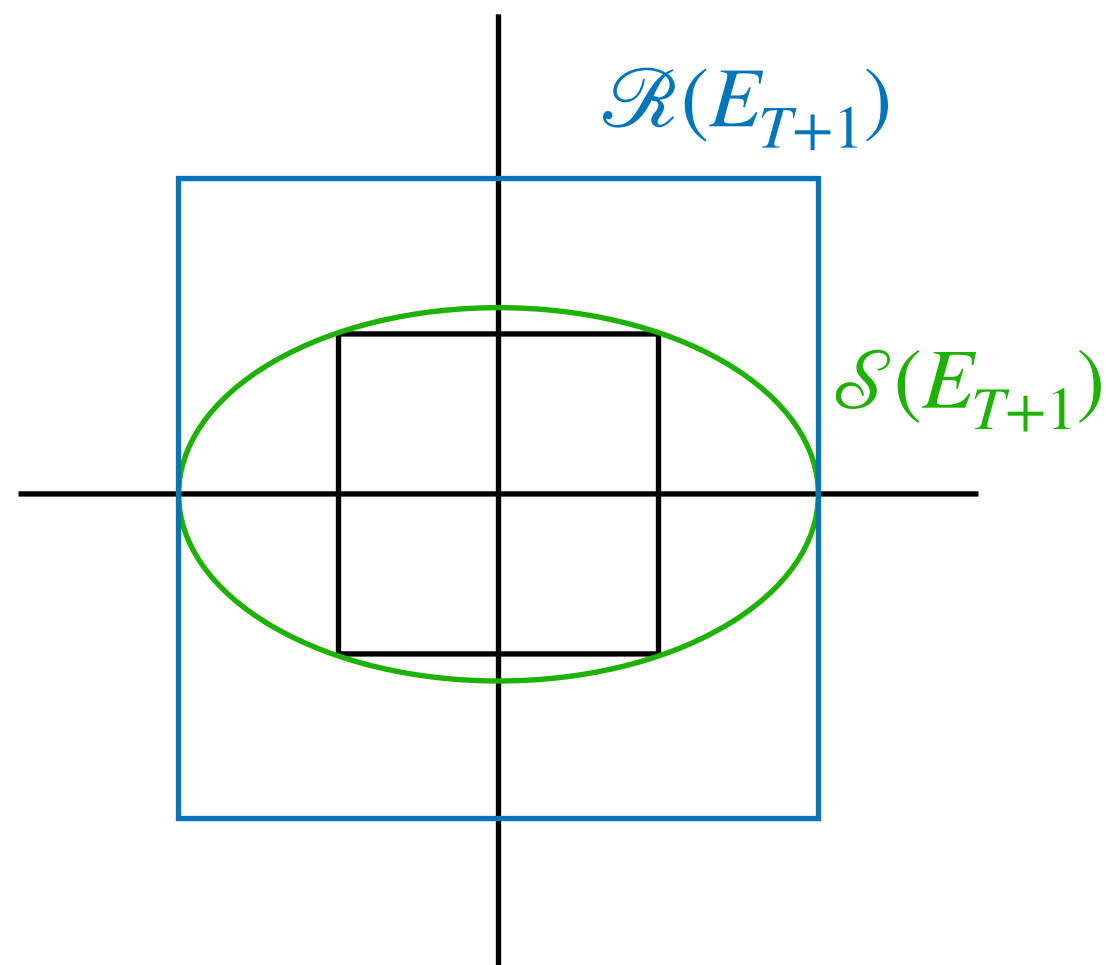
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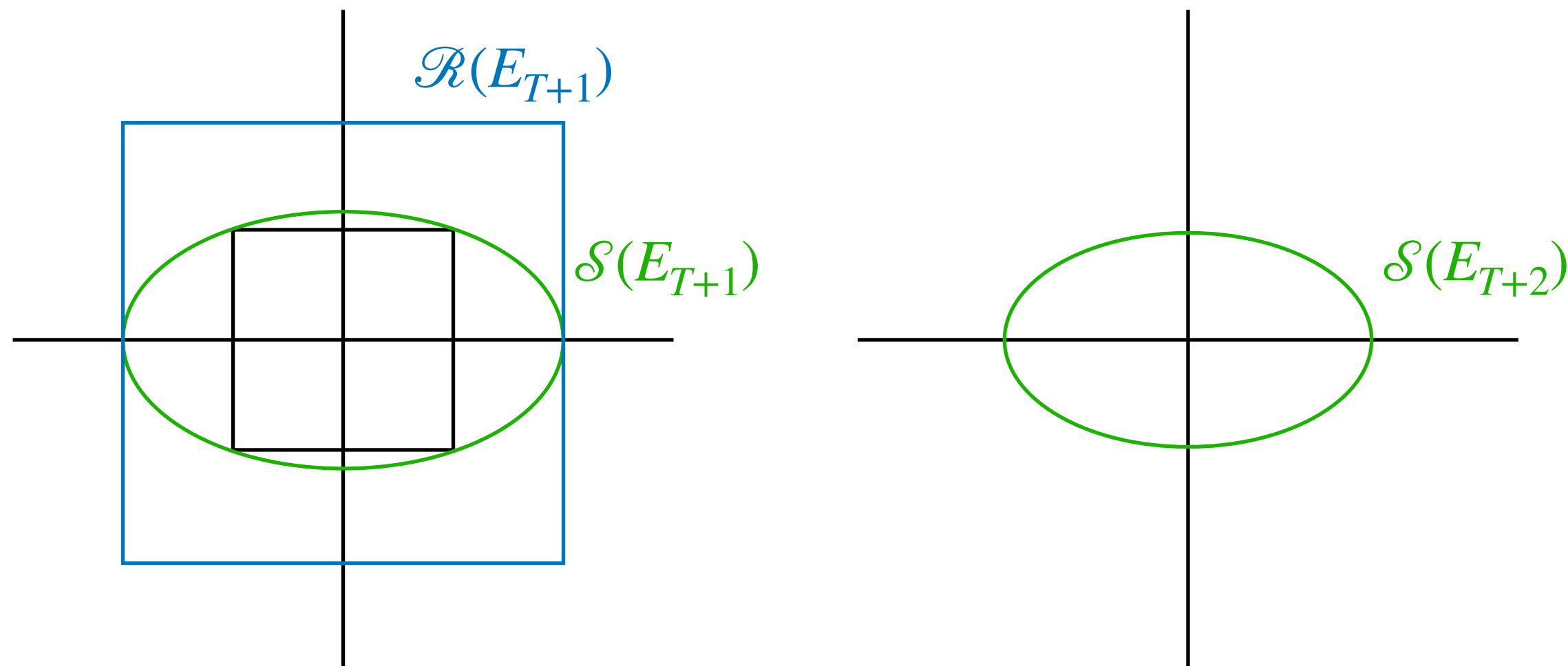
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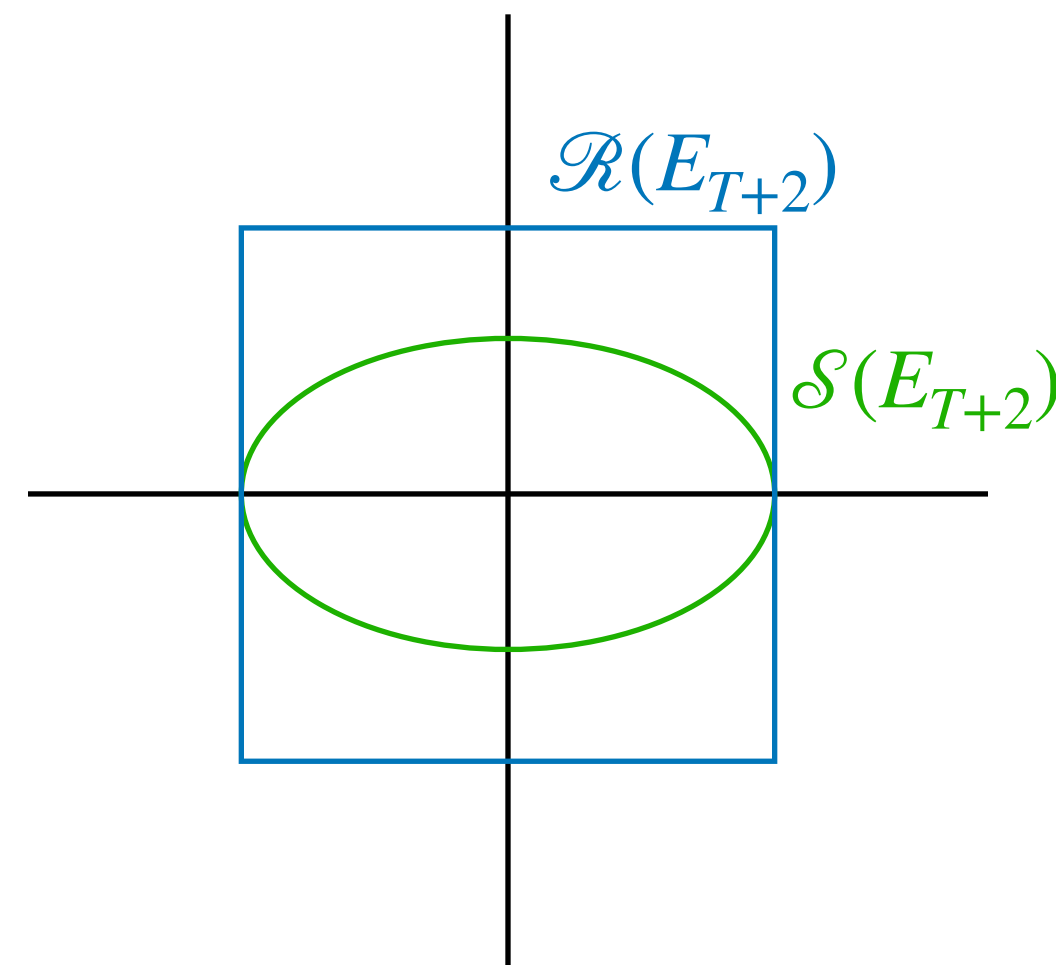
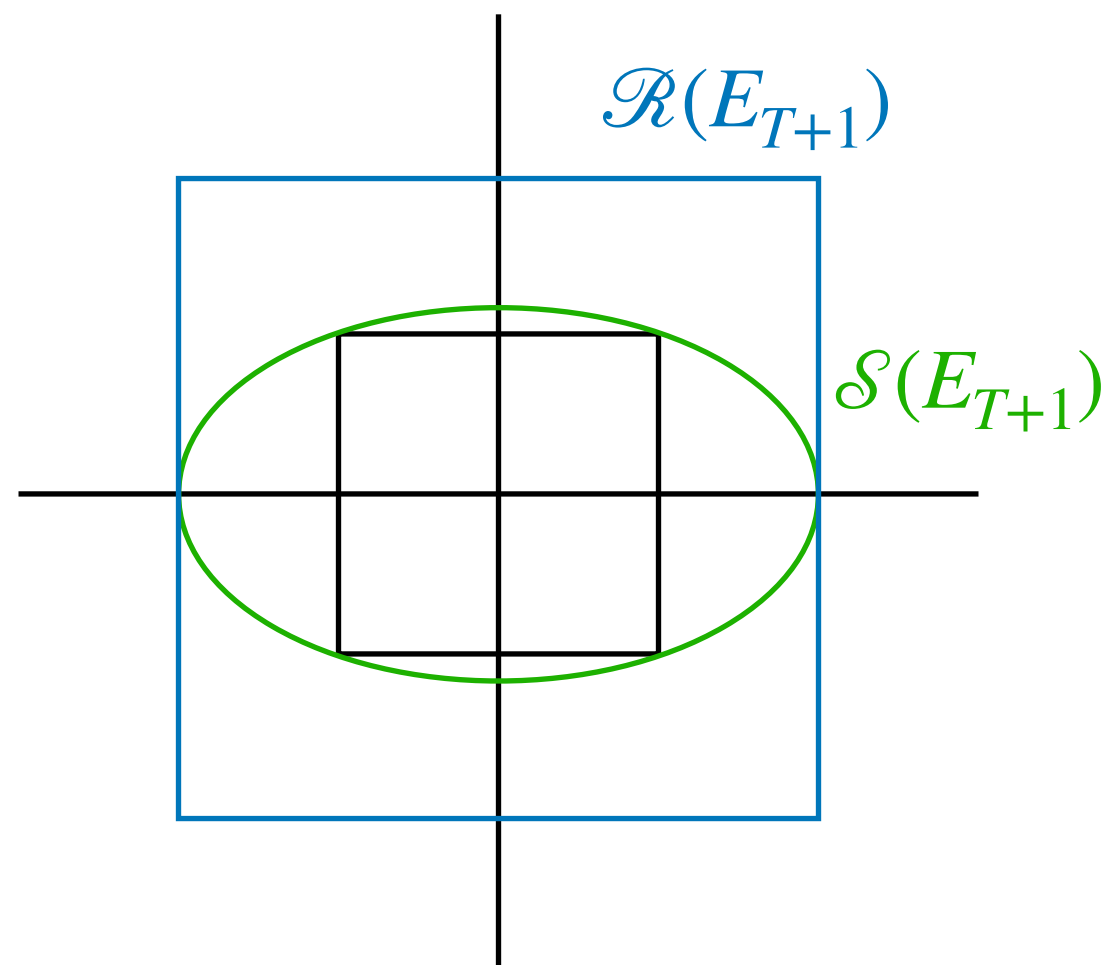
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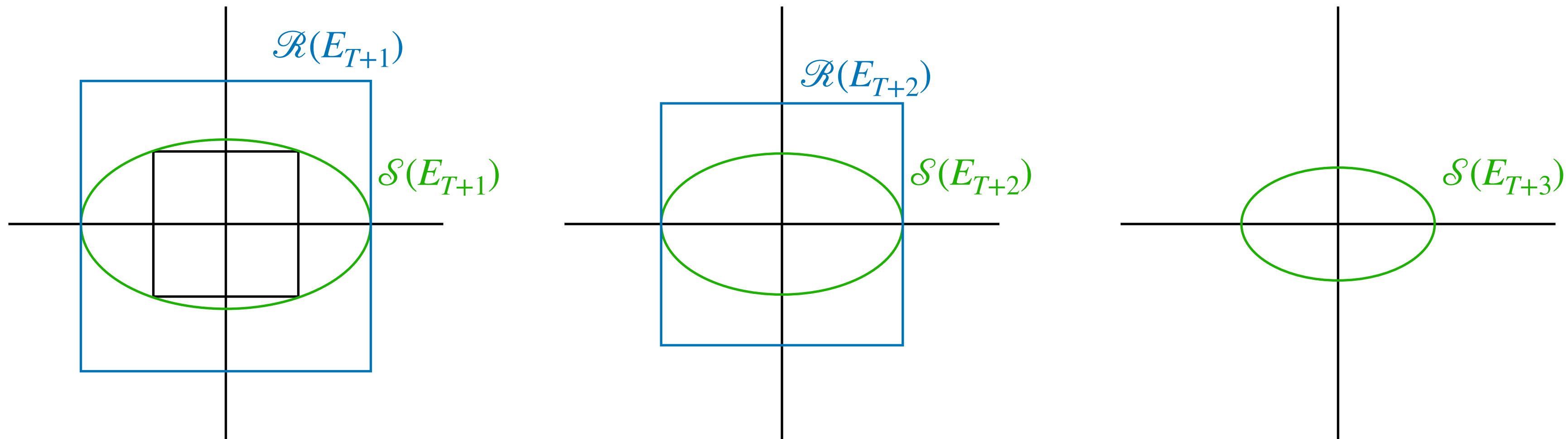
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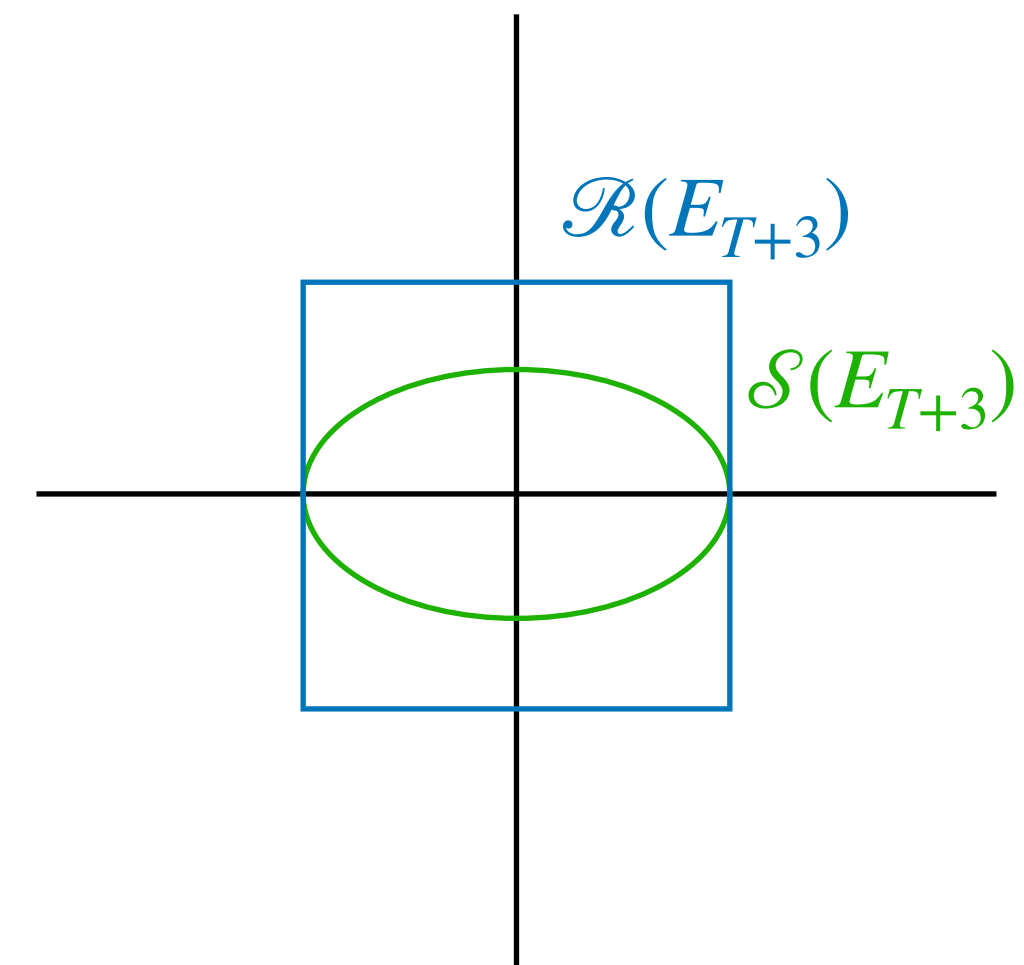
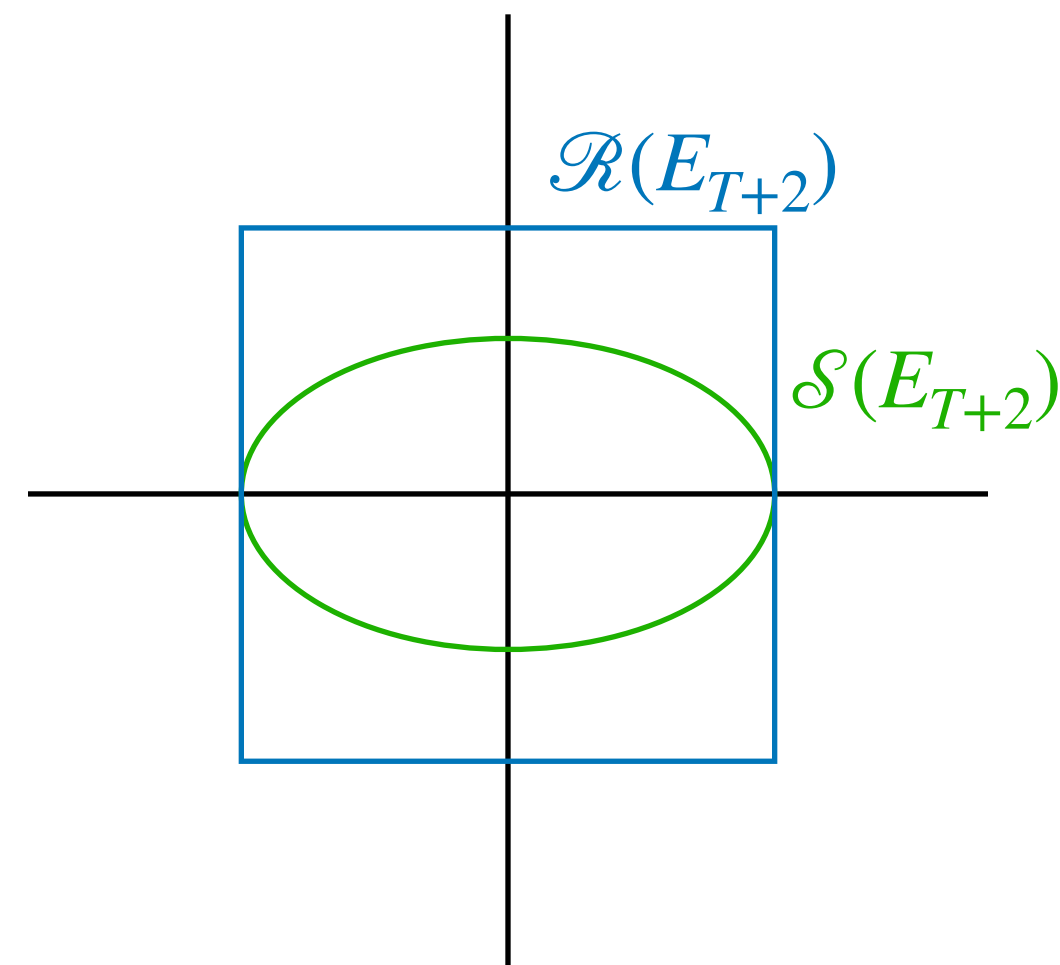
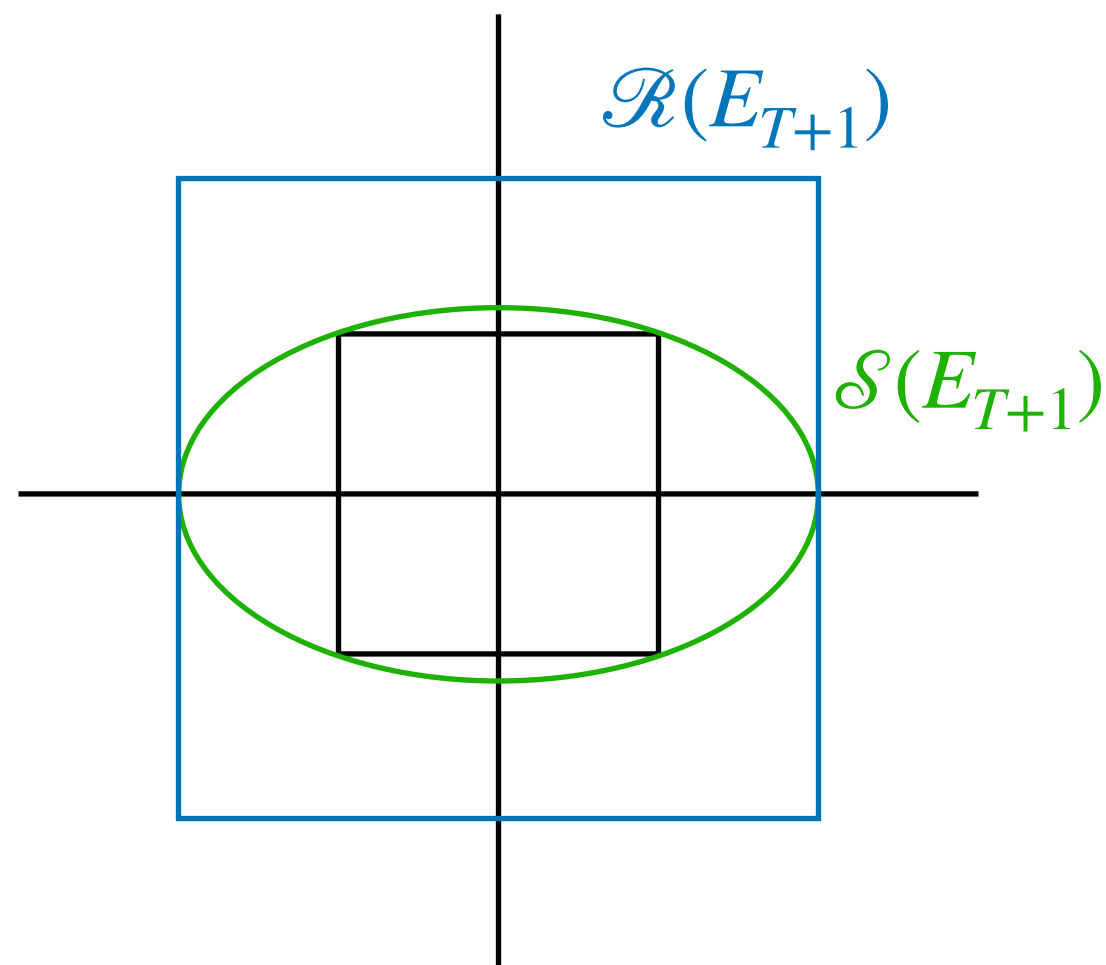
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Stabilization: Exponential convergence

Theorem 3 (Exponential convergence).

- Suppose

$$N_s > N^* := \frac{\theta + \sqrt{\theta^2 + \eta_1 \phi}}{\eta_1} \sqrt{\frac{n \bar{\lambda}(\bar{P})^3}{\underline{\lambda}(\bar{P})}},$$

where η_1 is the coefficient in the verifiable certificate.

- Then suitable contraction factor $\mu \in (0,1)$ can be selected such that

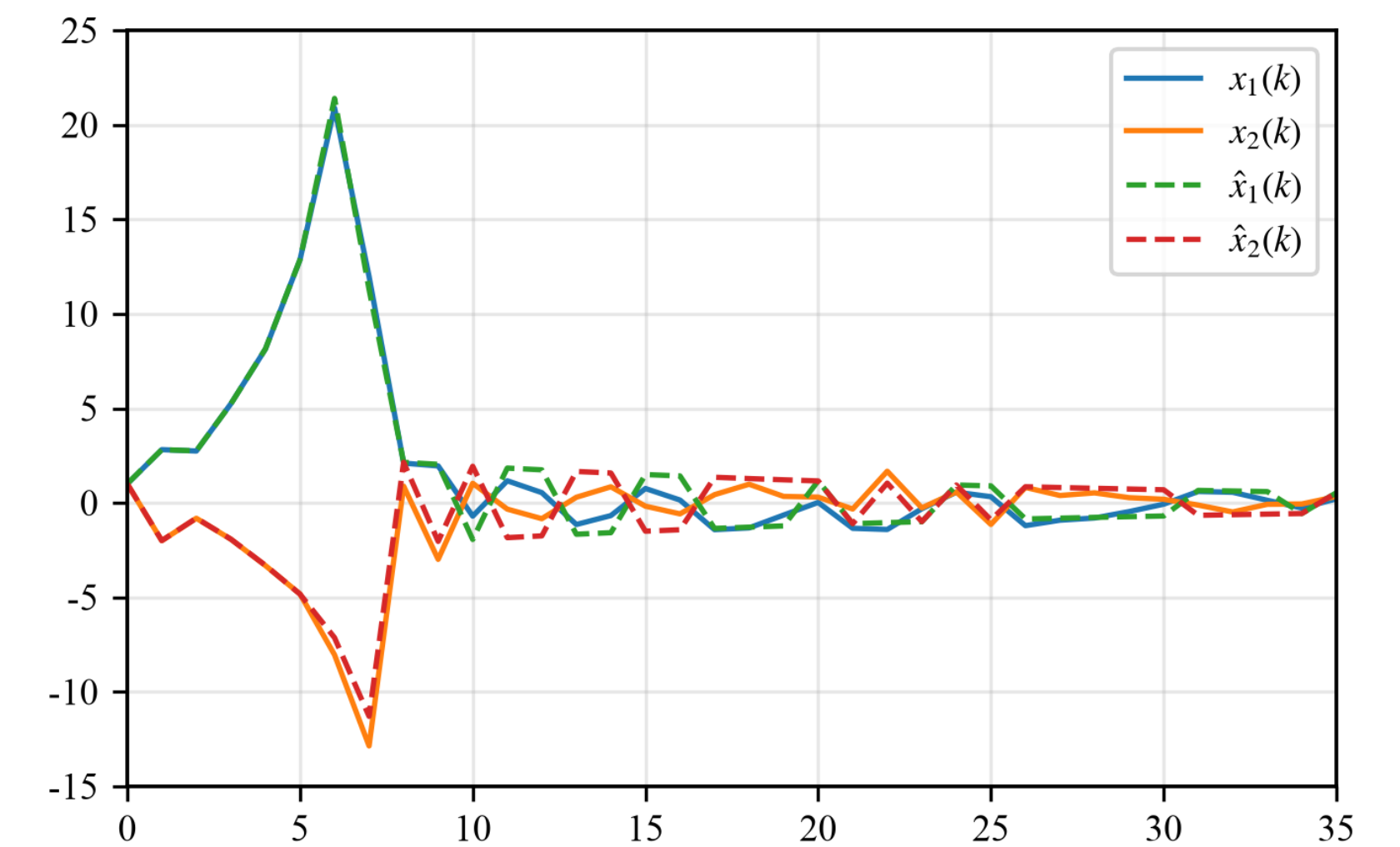
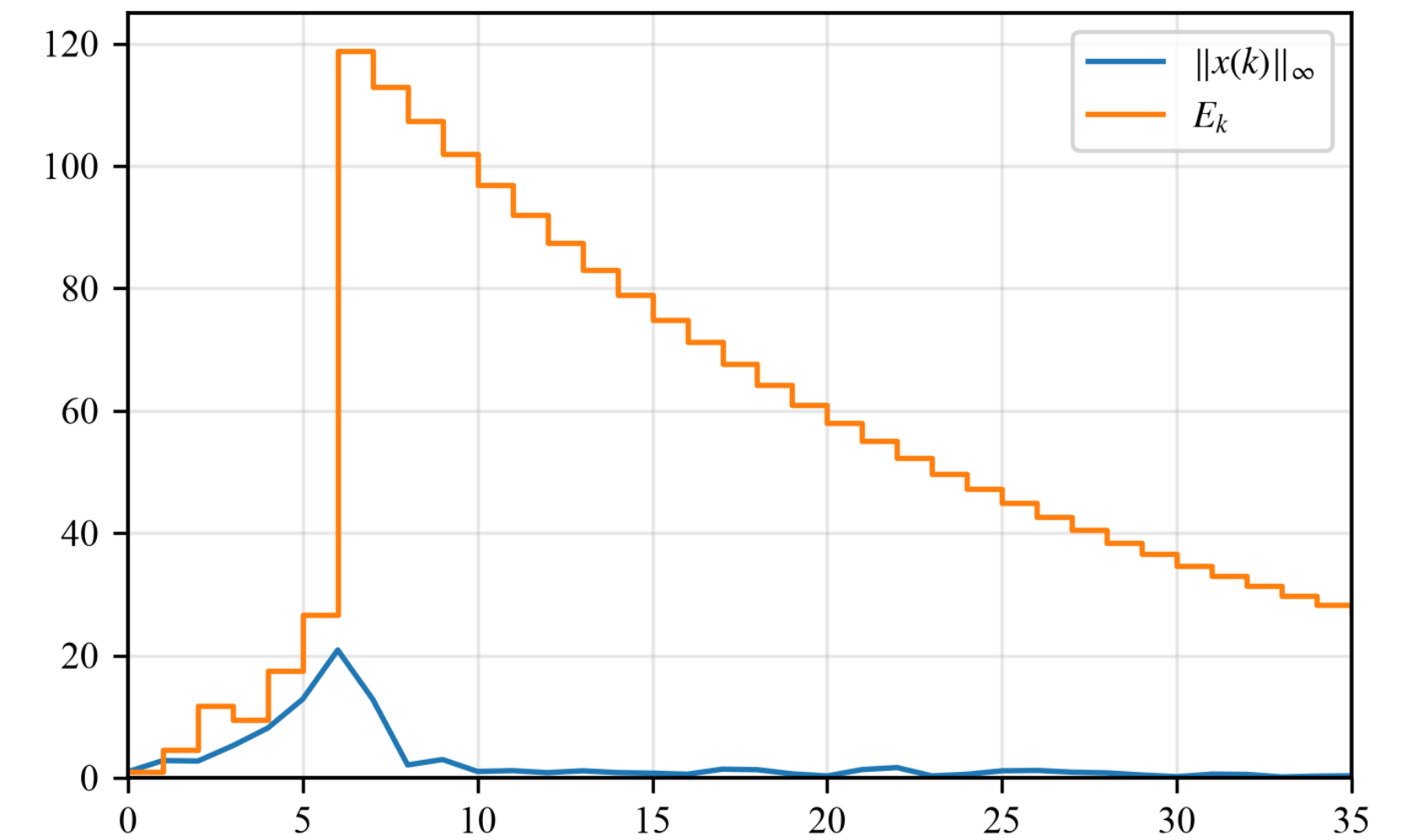
$$x(k) \in \mathcal{S}(E_k) \implies x(k+1) \in \mathcal{S}(\mu E_k),$$

and thus $x(k) \rightarrow 0$ exponentially.

Numerical illustration

LTI system with $A = \begin{bmatrix} 2 & 1 \\ -1 & -1 \end{bmatrix}$, $B = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$

- $T = 5$, $\delta = 0.01$
- $\bar{K} = [-1.129 \quad -0.333]$, $\eta_1 = 0.338$



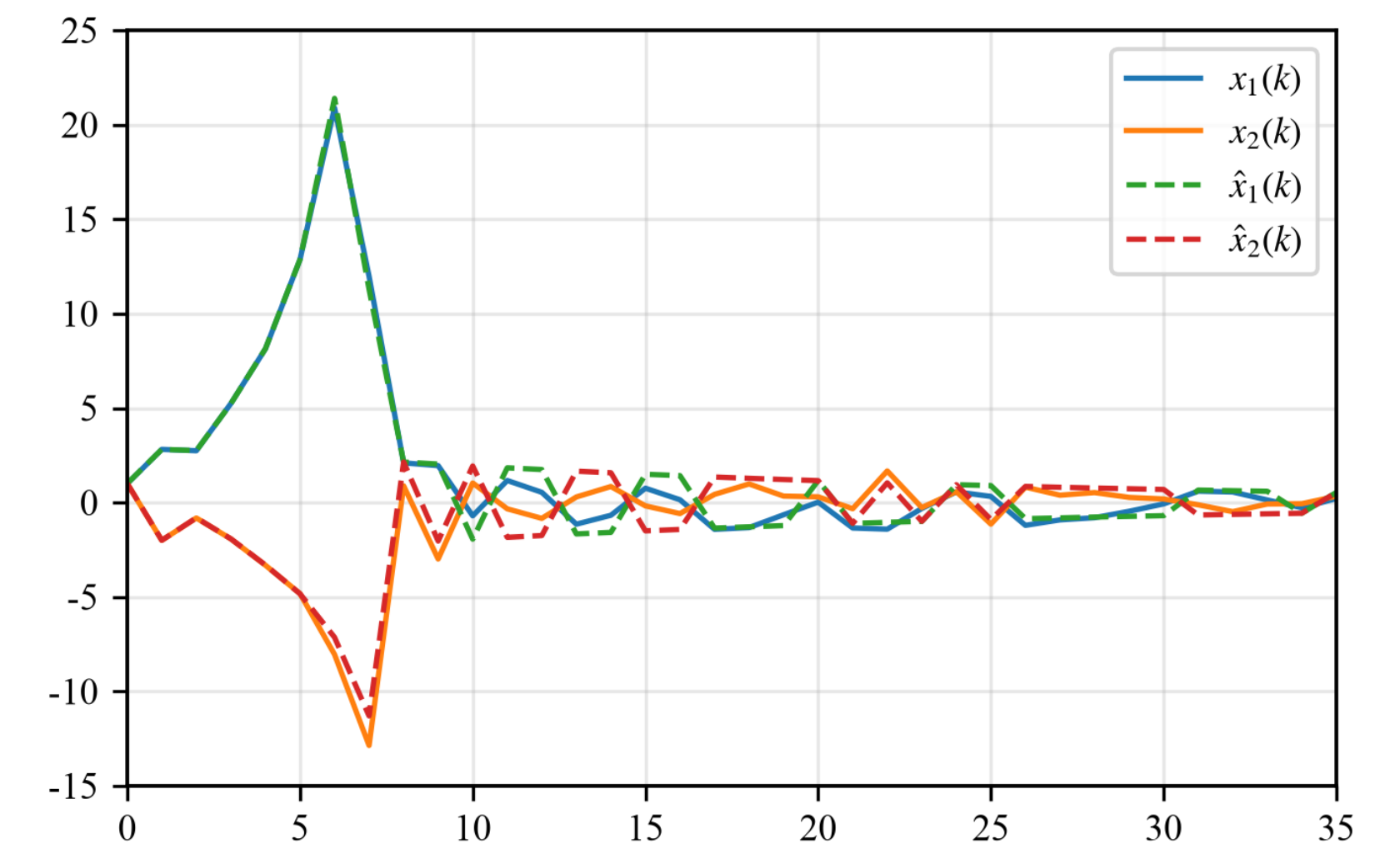
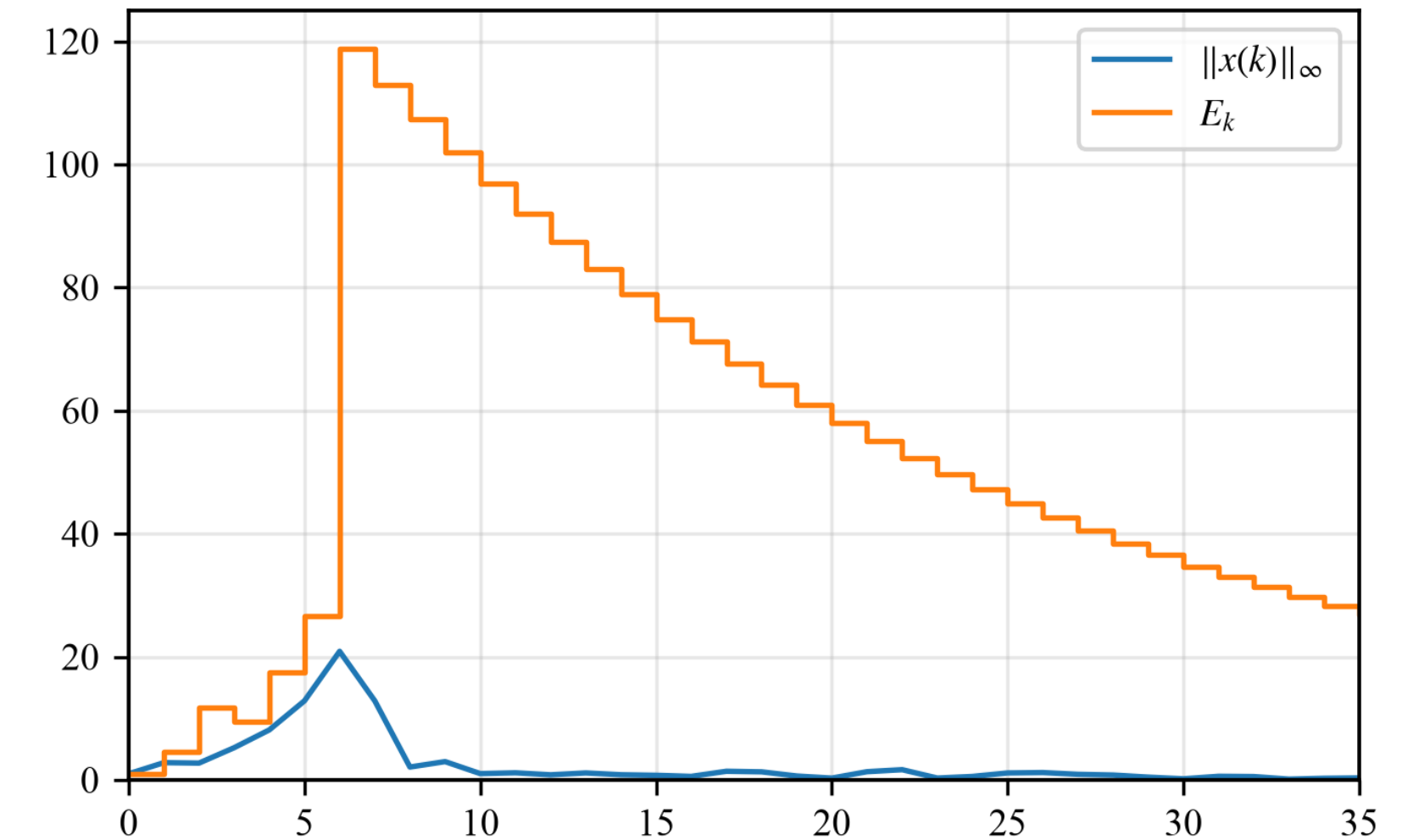
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Quantization:

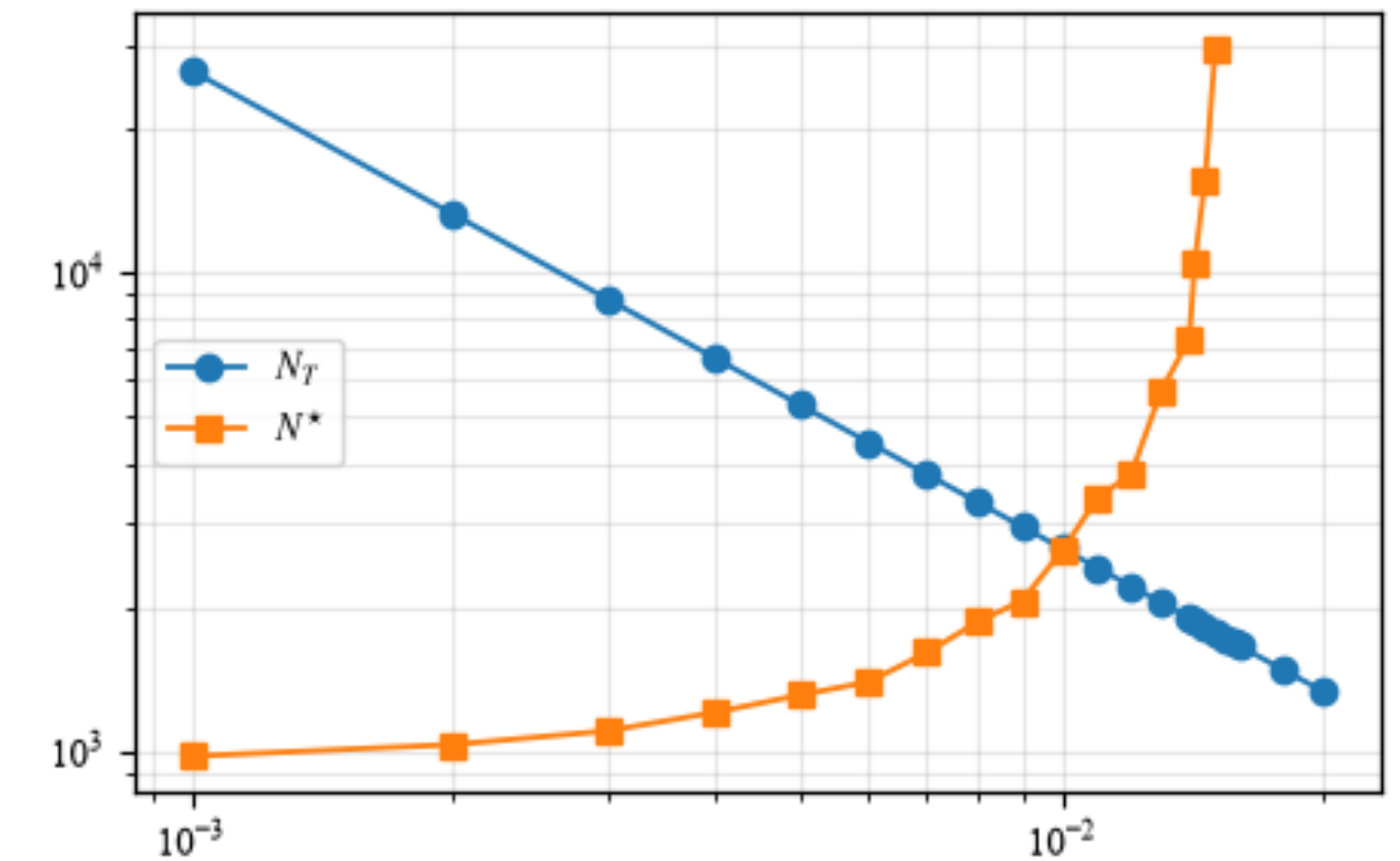
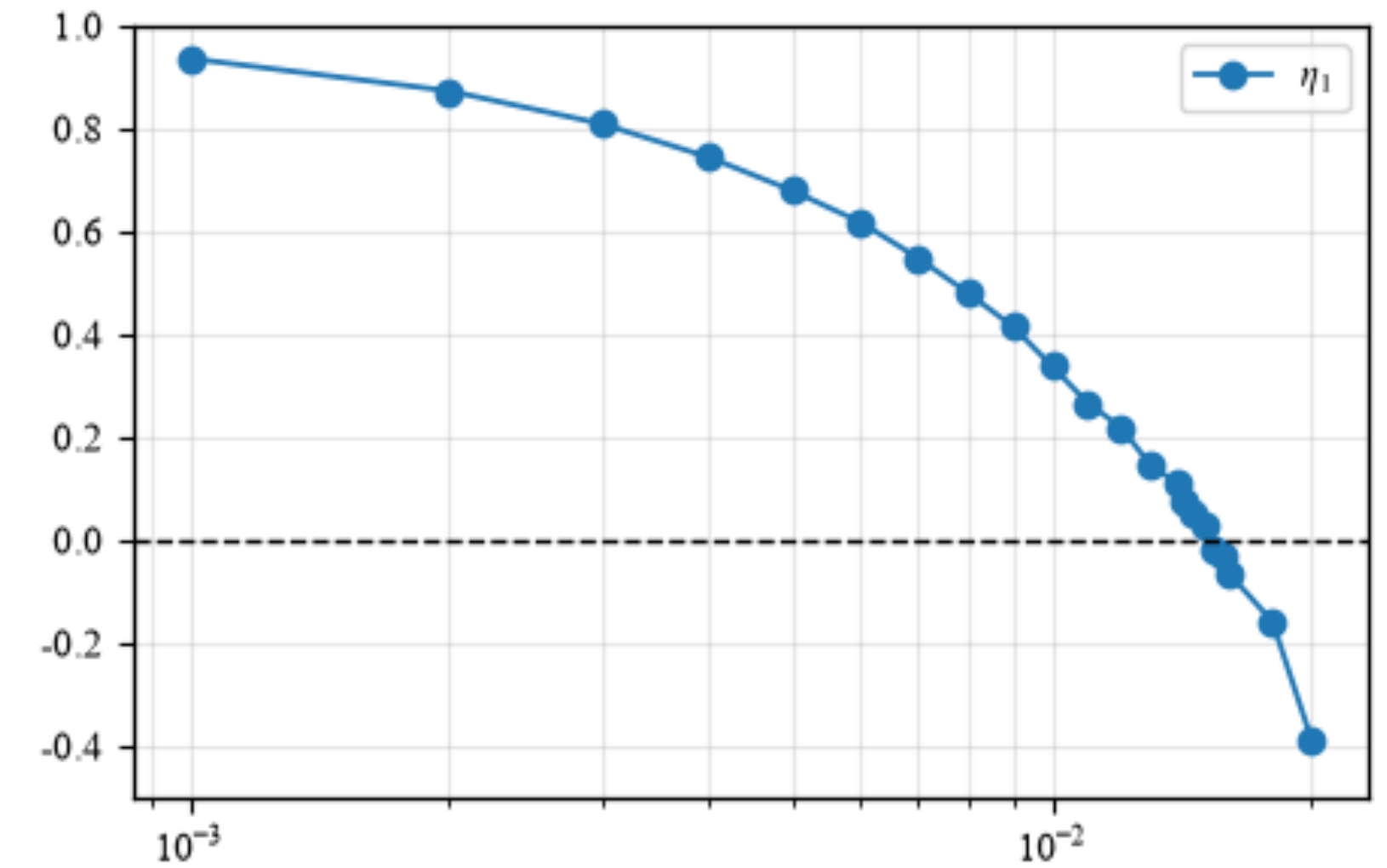
- **Controller design:**
Measurement-dependent update $\implies N_T : 42450 \rightarrow 2658$
- **Stabilization:**
Simulation: $N_s = 50$, $\mu = 0.95$
Theoretical sufficient bound: $N_s \geq 2624$



Quantization trade-offs

Proposition 1.

- Suppose controllable (A, B) and PE input.
- **As $\delta \rightarrow 0^+$:**
During controller design, $N_k = \Theta(\delta^{-1})$
For all optimal SDP solutions, $1 - \eta_1 = O(\delta)$ and $N^* = O(1)$



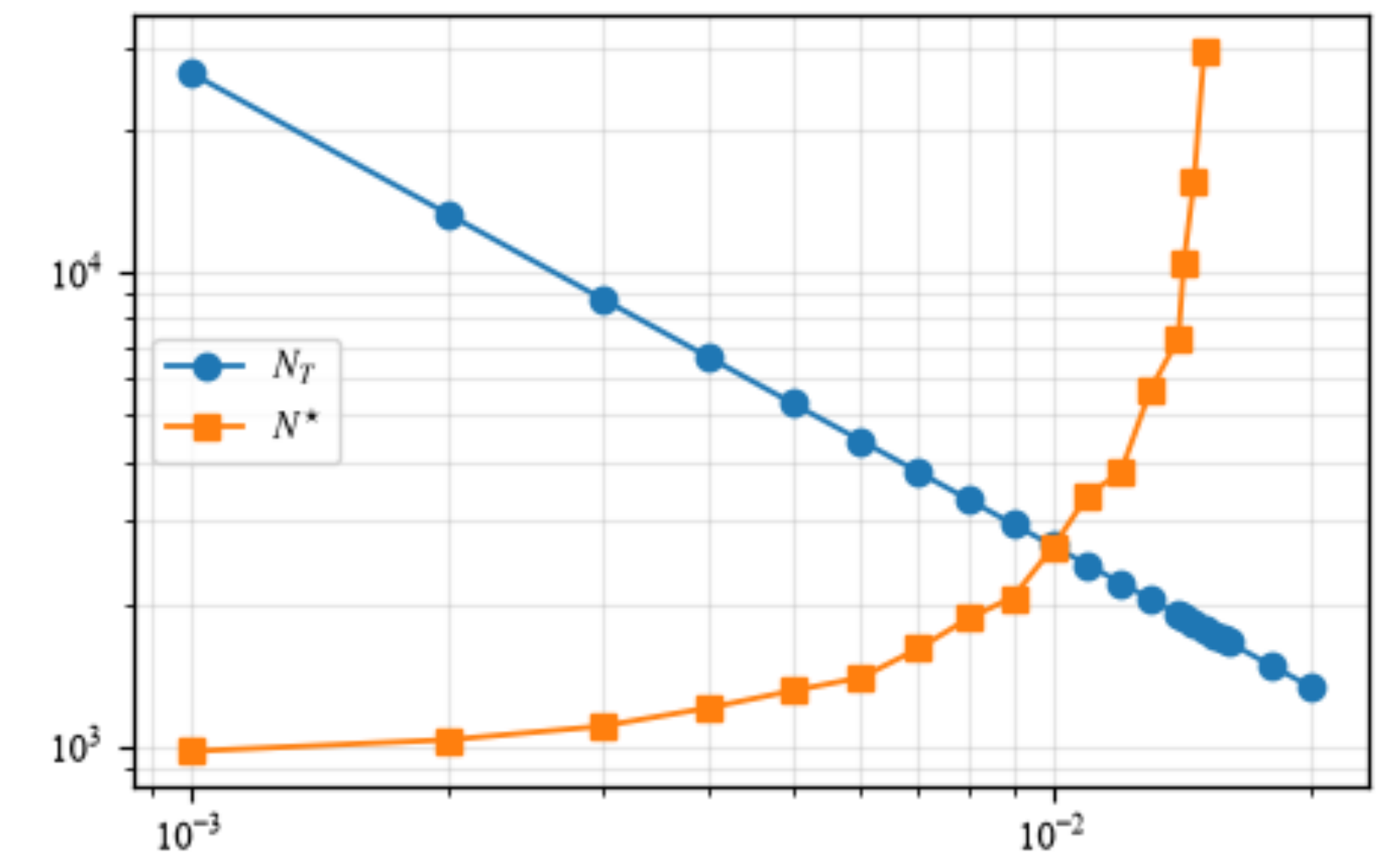
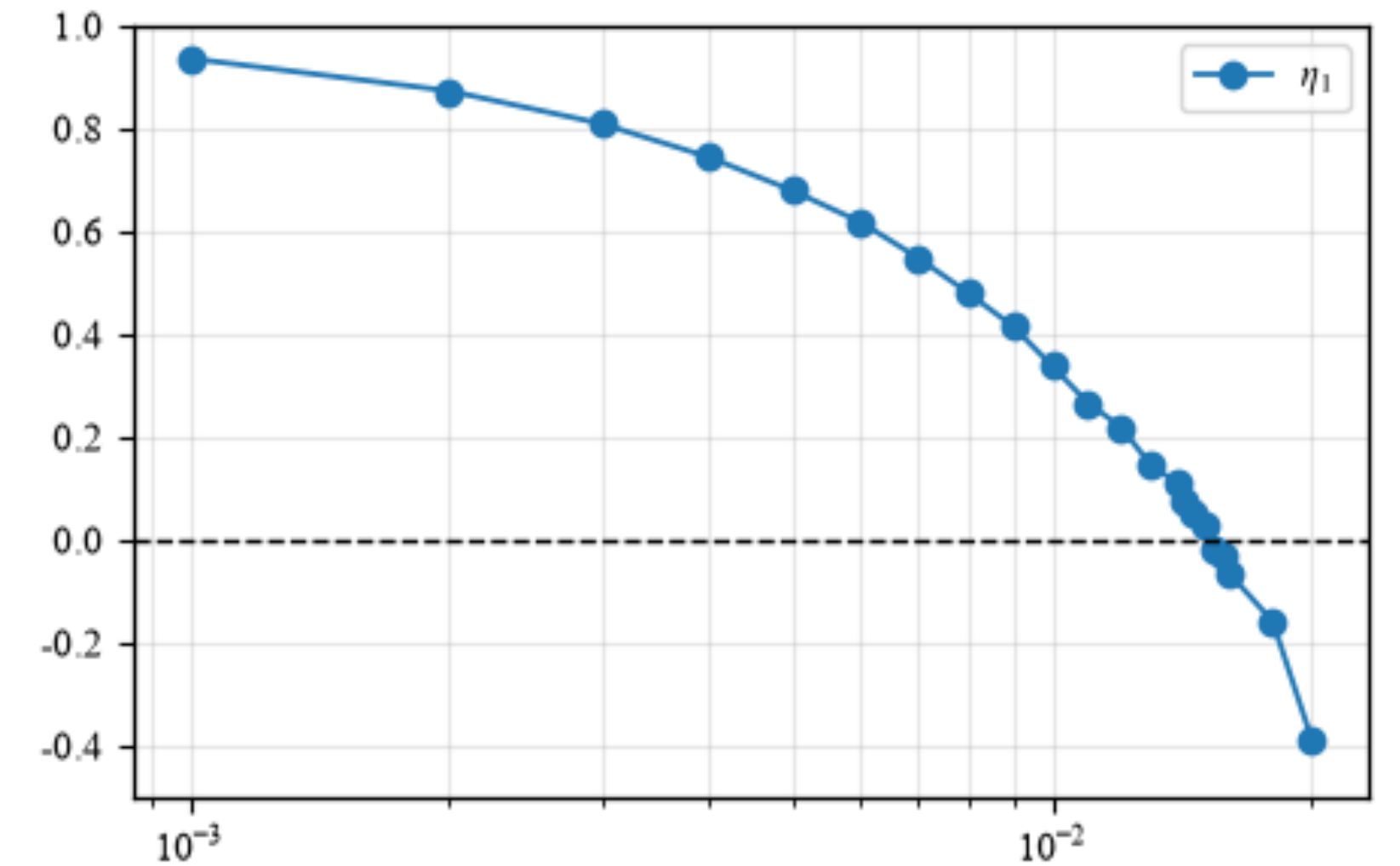
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Proposition 2.

- Suppose $\rho(A) > 1$.
- For all feasible SDP solutions such that $\eta_1 > 0$, **as $\eta_1 \rightarrow 0^+$:**
 $N^* = \Omega(\eta_1^{-1})$
- Under additional boundedness conditions, $N^* = \Theta(\eta_1^{-1})$



Conclusions

- **Controller design:**

Stabilizing feedback can be designed directly from finite-valued quantized state data.

- **Stabilization:**

A fixed finite number of quantization cells is sufficient for exponential convergence.

- **Quantization trade-off:**

Finer quantization during controller design requires more cells for data collection, but can reduce the sufficient quantization requirement for stabilization.

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Ongoing:

- Reduce conservativeness of the sufficient quantization bounds.
- Relax assumptions on known bounds for $\|x(0)\|$, $\|A\|$, and $\|B\|$.

Thank you!